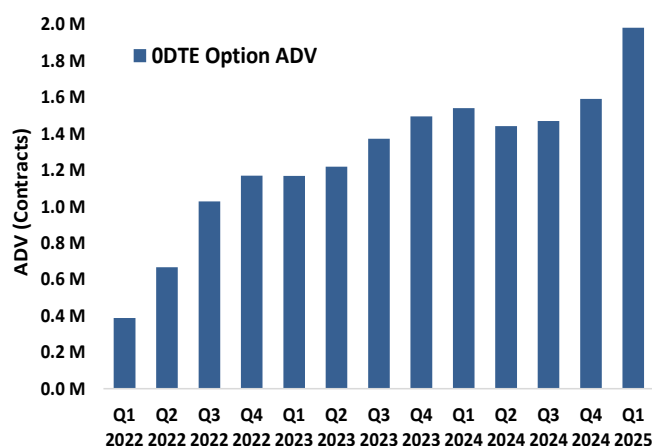


ODTEs Decoded: Positioning, Trends, and Market Impact

Retail and Institutional Trends in SPX® ODTE Options

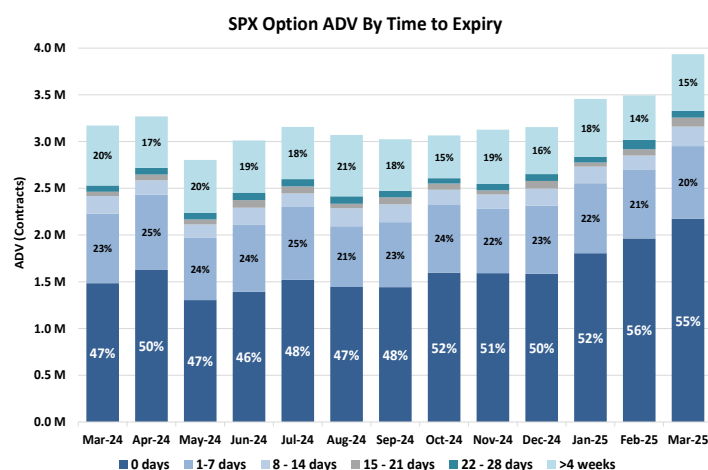
Since the introduction of Tue/Thu expiries in 2022, SPX zero day to expiration (ODTE) options trading have grown more than five-fold, with ADV jumping from 388k contracts in Q1 2022 to 1.98M contracts in Q1 2025 (Exhibit 1). ODTE options now make up over 50% of the total SPX volume, with over \$1T notional trading daily (Exhibit 2).

Exhibit 1: SPX ODTE Options Have Grown 5x in Last 3 Years



Source : Cboe

Exhibit 2: ODTE Options Now Over 50% of Total SPX Volume



Source : Cboe

What's driving the popularity of ODTE options? Increased utility is certainly one factor, as daily expiries allow traders the ability to manage risk more precisely. The low dollar premium of ODTE options make these a cheap way to express directional/tactical views while the high volatility risk premium embedded in these ultra-short dated options have made it an attractive target for yield harvesters/income seekers.

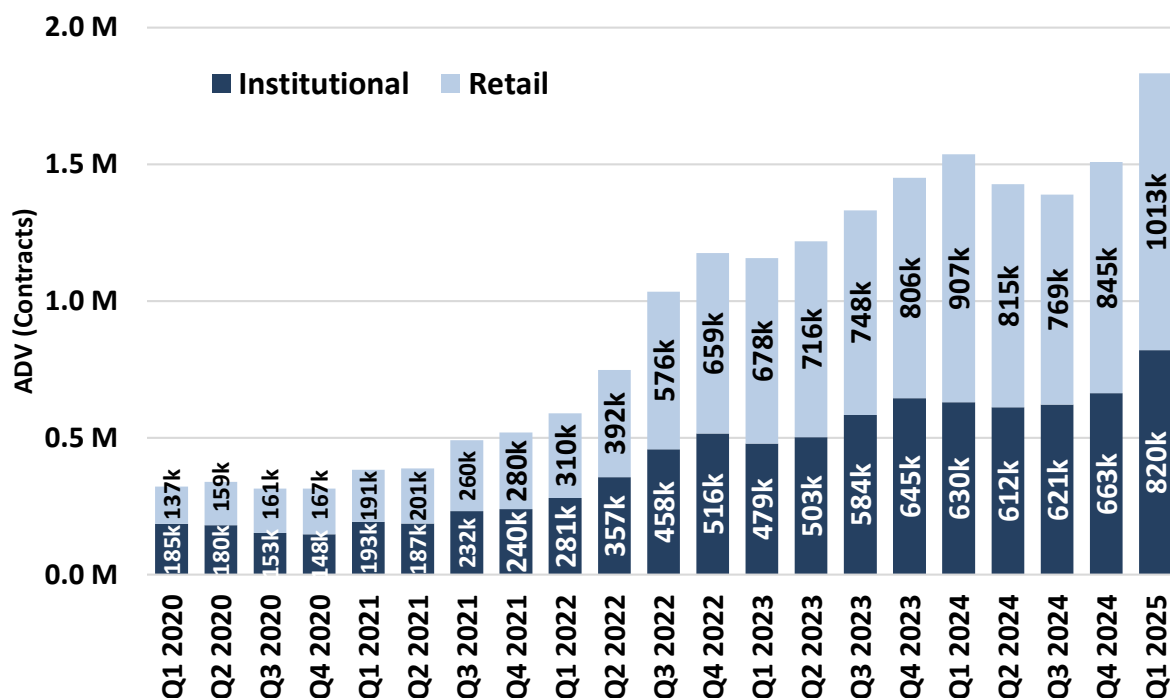
Beyond that, wider adoption is another big driver. Retail options trading has surged since the pandemic on the back of zero commissions, increased education, and wider access, with Robinhood the latest retail broker to enable index options trading in January this year.

Rise of Retail

This leads to the inevitable question – just how big is retail in SPX ODTE options? This turns out to be a surprisingly difficult question to answer. That's because while we as an exchange can see where an order originates from – about 90% of SPX ODTE customer volume currently originate from a retail brokerage firm – that is not sufficient to identify whether a trade is indeed retail or institutional in nature. That's because many institutional investors also use some of these retail platforms for execution (e.g. Interactive Brokers).

Instead, we can make some educated guesses on the breakdown of retail vs. institutional trading by looking at not just point of origin of trades, but also size of order, frequency of trading, risk exposure, as well as other proprietary exchange data. For example, someone trading a 1-lot is more likely to be retail than institutional (note: we look at order size, not execution size...this way even if a large order is broken down and executed in smaller lots, it would still be identified as an institutional order). We further qualify trade size by adding a risk component – the capital required to buy one SPX ODTE option is small (i.e. just the premium paid) while the capital required to naked short one SPX ODTE option is significant (i.e. 1 contract has notional size of ~\$550k). In our analysis, we assume that a trade with a high max loss/capital requirement is more likely to be institutional. Admittedly, these are just assumptions, and the line gets even more blurred as retail investors get more sophisticated. However, using the above criteria, we estimate that retail currently make up ~50-60% of SPX ODTE trading, with institutional (and more capitalized retail traders) making up the remaining 40-50%.

Exhibit 3: Estimated Retail vs. Institutional Volume in SPX ODTE Options



Source: Cboe

Using our estimates, we can clearly see the growth of retail SPX ODTE trading. Total retail ADV has surged over 7x from 137k contracts in Q1 2020 to over 1M contracts in Q1 2025 while institutional volumes have grown 4x from 185k to 820k during the same period (see Exhibit 3). Retail made up over 55% of SPX ODTE trading in Q1 this year compared to 42% five years ago.

Retail vs. Institutional Trading Characteristics

What are retail and institutional investors trading in ODTE options? Turns out, both deploy similar strategies. Just over half – around 50-52% – of opening customer trades for both groups are in outright puts and calls, with the rest in multi-leg trades. Institutional investors trade slightly more vertical spreads (33% vs. 28%) while retail is more active in iron condors and butterflies.

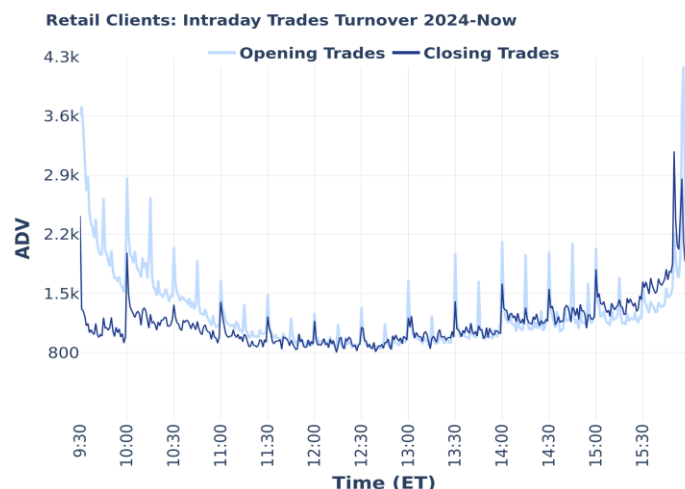
Exhibit 4: ODTE Strategy Breakdown

	% Total ADV	
	Institutional	Retail
Outright Put/Call	50.2%	51.9%
Vertical Spread	33.4%	27.6%
Iron Condor	4.1%	7.7%
Butterfly Spread	1.0%	3.5%
Straddle/Strangle	1.1%	0.3%
Others	10.2%	9.0%

Source: Cboe

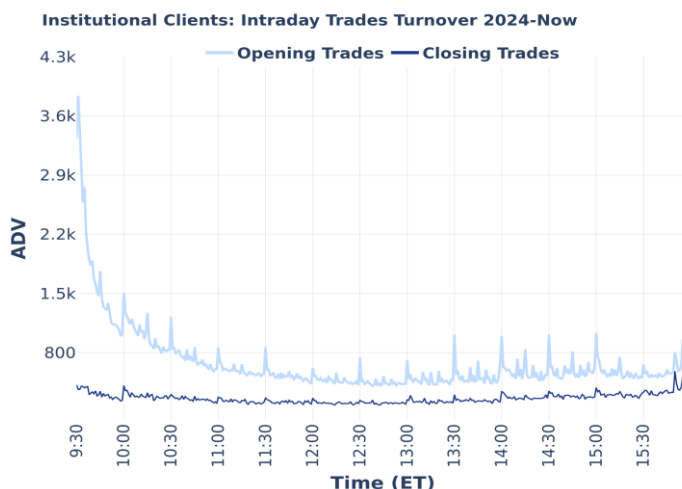
How they trade, however, can be quite different. When we examine the intraday trading patterns of retail vs. institutional, we can see that institutional investors are much more aggressive in opening positions at the beginning of the day with 18% of all opening trades happening in the first 30 minutes. Institutional opening activity trails off significantly after that. In contrast, retail investors are active both at the start as well as the end of the day, with ~12% of all opening trades occurring in the first 30 and last 30 minutes of the day. The U-shaped volume curve for retail ODTE activity very much resembles the typical intraday volume curve for equities. See Exhibit 5 and 6 below.

Exhibit 5: Retail Intraday Volume Profile



Source : Cboe

Exhibit 6: Institutional Intraday Volume Profile



Source : Cboe

When it comes to closing trades, there is also a sharp contrast. While institutional closing activity is fairly constant throughout the day and meaningfully lower than the opening activity, retail closing volume resembles the same U-shape curve as its opening volume (with the two lines overlapping for much of the day). This suggests that retail is much more active initiating and unwinding positions throughout the day, while institutional investors tend to leave positions open longer due to either higher risk tolerance, or because they have other ways to hedge open positions (e.g. delta-hedging or offsetting positions elsewhere).

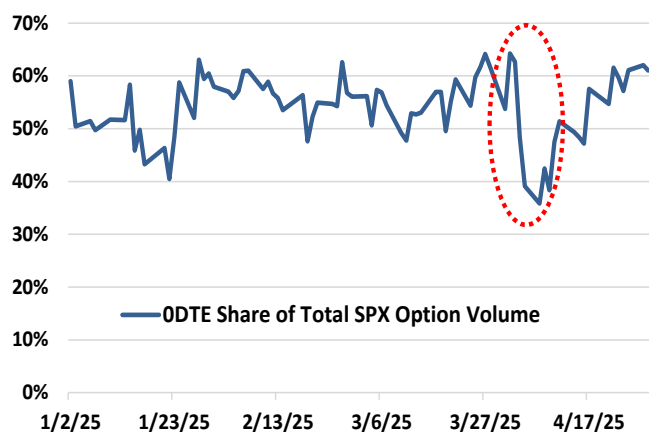
Another way the two groups differ is in their risk tolerance. Over half – 52% to be exact – of all retail opening trades have a max loss of under \$2k. That is especially impressive given the notional size of one SPX contract is ~\$550k. In contrast, 61% of institutional opening volume are in trades with max loss of over \$20k, with 27% in trades with max loss of over \$100k. Retail traders are able to keep a very tight leash on their risk exposure through either outright long options (with max loss capped at the premium paid), or by selling options through tight spreads. For example, the most commonly traded vertical spread width, at 34% share, is just 5 point wide (i.e. \$500 max loss).

Taken together, we can see retail traders have multiple ways of managing their risk – by closing positions quickly if they start to go against them, capping risk at the outset through tight spreads or outright long options, or as we'll see in the next section, by staying on the sidelines during sudden spikes in volatility.

Navigating Recent Tariff-Induced Volatility

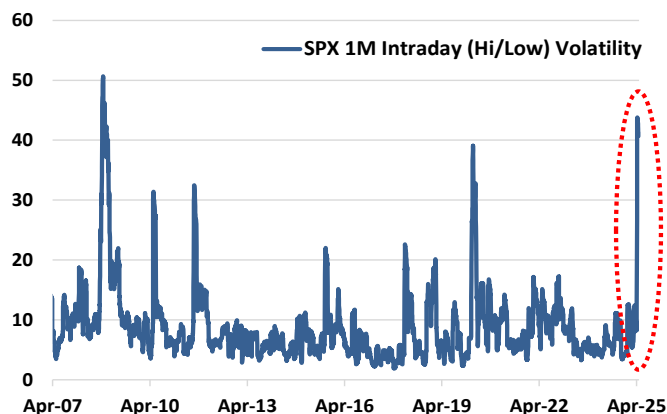
While overall SPX option volumes surged during the recent market turmoil, the share of ODTE options traded went down notably. ODTE share of overall SPX option volume fell from a high of 64% on April 1st to a low of 36% on April 7th (Exhibit 7) when the VIX® index hit a high of 60.

Exhibit 7: ODTE Share Fell to 36% During Recent Turmoil



Source : Cboe

Exhibit 8: Intraday Volatility Highest Since 2008 GFC



Source : Cboe

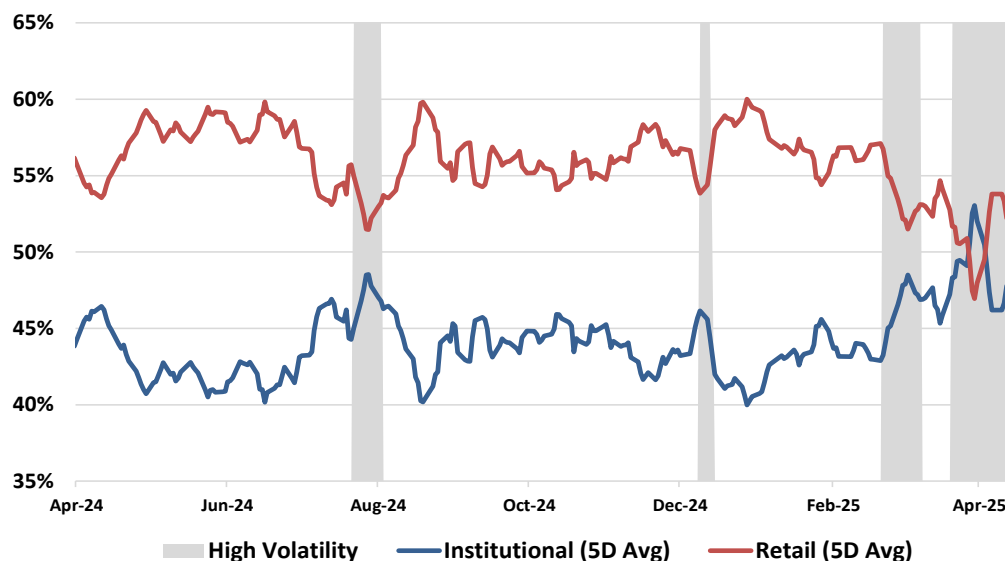
If we zero in on the three days when the ODTE share fell below 40% (April 4th, 7th, and 9th), we can see that while ODTE volumes did go down, the average volume on those three days was only 10% lower than the previous 20-day

average, at 1.97M contracts per day (vs 20-day average of 2.18M contracts). What drove the sharp decline in the %share was the fact that longer-dated options saw a huge uptick in volume. Options expiring in 4+ weeks experienced the biggest increase, up 141% from 618k ADV to 1.49M contracts on those three days. Trump's tariffs created massive uncertainty around the longer-term outlook of US inflation and growth, and it's no surprise that investors gravitated toward longer-dated options to reposition their portfolios accordingly. With volatility abating in the past two weeks on the back of the 90-day pause in reciprocal tariffs, ODTE options have resumed its dominant share of overall SPX volume, now back to a high of 60%.

When we dig a little deeper into the drivers of recent ODTE flow, we see a notable pullback by retail investors in ODTE trading (see Exhibit 9), with the retail share of ODTE volume dropping 10 pts from 57% in March to 47% on April 9th (the day SPX swung over 10% intraday). This is on the back of two factors – the jump in implied volatility making outright options more expensive and the even bigger jump in realized volatility making short option strategies less profitable. As shown in Exhibit 8 above, SPX intraday volatility surged this month to its highest level since the 2008 Global Financial Crisis, higher than even Covid (when the VIX index hit 80). It's no surprise retail investors pulled back in the face of historic intraday volatility.

Exhibit 9: Retail Traders Have Pulled Back Amidst Recent Volatility

Estimated Retail vs. Institutional Share of SPX ODTE Trading



Source: Cboe

In fact, as we can see above, it's not unusual for retail to pull back in the face of elevated volatility (such as last August 5th), but such drops are usually temporary, with retail activity picking up once again when volatility moderates. In their place, institutional investors step in, taking on a greater share of the volume. In addition to their higher risk tolerance, institutional traders often use ODTE options during periods of high volatility to help manage risk

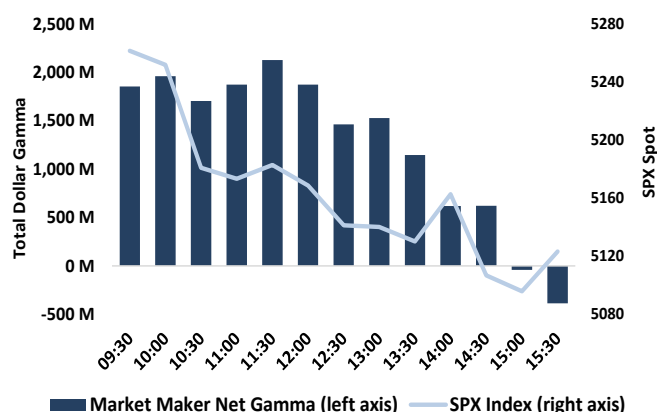
in the rest of their portfolio as they provide a cost-effective way to trade delta or hedge gap risk for the overall portfolio.

While ODTE volumes may ebb and flow in the short term, longer term it has continued to grow as more investors are able to access its utility. We see the resilience of ODTE options trading as partly due to the prudent risk management of ODTE traders, with over 95% of all ODTE trades done in capped risk formats (i.e. either long options outright or short options through spreads). Naked short options, with the potential for massive losses, make up just 4% of SPX ODTE trading.

Could ODTE Options Be Behind the Recent Market Volatility?

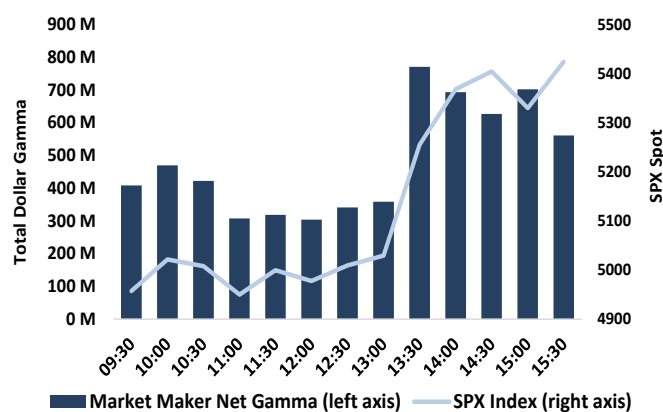
As usual, when market volatility picks up, there is a [chorus of voices](#) pointing to ODTE options as a potential trigger. Given the size of SPX ODTE trading, now totaling over \$1T notional a day, the fear is that these options could potentially exacerbate swings in the underlying SPX index due to outsized market maker hedging activity. But as we laid out extensively in our [previous piece](#) (*“Much Ado About ODTEs: Evaluating the Market Impact of ODTE Options”*), that fear is misplaced. High volume doesn’t equal high risk. What matters for determining the potential impact of market maker gamma hedging activity is the *balance* of the volume between buys vs. sells, not the notional size. And what’s remarkable about SPX ODTE flow is how balanced it is between buyers and sellers, puts and calls. As we outlined above, both institutional and retail investors use these options for a range of purposes – from tactical bets to systematic yield harvesting. This is why the put/call ratio for SPX ODTE options have consistently hovered around one, in sharp contrast to non-ODTE options (where the primary use case is hedging). This is also why the net gamma exposure (or market maker positioning) of ODTE options have been so minimal.

Exhibit 10: Market Maker Positioning on Apr 4th



Source : Cboe

Exhibit 11: Market Maker Positioning on Apr 9th



Source : Cboe

We can see this by looking at two particularly volatile days recently – April 4th when SPX fell 6% and April 9th when SPX surged 10%. As you can see in Exhibit 10, net market maker gamma exposure in SPX ODTE options ranged from

+\$2.1B to -\$390M on April 4th with market makers mostly long gamma throughout the day (i.e. they were hedging in the *opposite direction* of the market move, so if anything, they were dampening volatility, not exacerbating it). Similarly, market makers were net long gamma throughout April 9th, ranging from +\$300M to +\$770M (Exhibit 11). While these numbers may seem big, it's important to put them in the context of overall SPX liquidity. SPX futures traded over \$850B on April 9th and over \$940B on April 4th, so market maker gamma hedging activity in SPX ODTE options represented, at best, just 0.2% of S&P futures notional volume.

Hardly the tail wagging the dog.

Acknowledgements:

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